

PRIMERA SUBIDA DE UN SERVICIO EN LA NUBE



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Arquitectura en la nube
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WSL

Parte 1: Apache + PHP

Primero actualizaremos los paquetes del sistema con los comandos “***sudo apt update && sudo apt upgrade -y***”.

```
lumno13@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG$ sudo su
[sudo] password for lumno13:
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
```

```
41 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# sudo apt upgrade -y
Reading package lists... Done
```

Para instalar el servidor web Apache usaremos “***sudo apt install apache2 -y***”.

```
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap
  libaprutil1t64 liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser ufw
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap
  libaprutil1t64 liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.
Need to get 2086 kB of archives.
After this operation, 8090 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [1
08 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB
]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7
[11.2 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9116
B]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.8 [1
331 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.8 [16
3 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.8
[97.7 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.8 [90.2
kB]
```

Para instalar el PHP y su módulo para Apache, ***“sudo apt install php libapache2-mod-php -y”***.

```
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# sudo apt install php libapache2-mod-php -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libapache2-mod-php8.3 php-common php8.3 php8.3-cli php8.3-common php8.3-opcache php8.3-readline
Suggested packages:
  php-pear
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.3 php php-common php8.3 php8.3-cli php8.3-common
  php8.3-opcache php8.3-readline
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 4922 kB of archives.
After this operation, 22.4 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 php-common all 2:93ubuntu2 [13.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-common amd64 8.3.6-0ubuntu0.24.04.5 [740 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-opcache amd64 8.3.6-0ubuntu0.24.04.5 [371 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-readline amd64 8.3.6-0ubuntu0.24.04.5 [13.5 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-cli amd64 8.3.6-0ubuntu0.24.04.5 [1915 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libapache2-mod-php8.3 amd64 8.3.6-0ubuntu0.24.04.5 [1851 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 libapache2-mod-php all 2:8.3+93ubuntu2 [4224 B]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3 all 8.3.6-0ubuntu0.24.04.5 [9174 B]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 php all 2:8.3+93ubuntu2 [4076 B]
Fetched 4922 kB in 0s (14.0 MB/s)
Selecting previously unselected package php-common.
(Reading database ... 41489 files and directories currently installed.)
Preparing to unpack .../0-php-common_2%3a93ubuntu2_all.deb ...
```

Iniciaremos el servicio de Apache, y comprobaremos su funcionamiento con los siguientes comandos respectivamente: ***“sudo service apache2 start”*** y ***“sudo systemctl status apache2”***.

```
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# sudo service apache2 start
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-10-03 09:02:31 CEST; 53s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 9961 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 9964 (apache2)
    Tasks: 6 (limit: 9350)
   Memory: 10.5M (peak: 11.9M)
      CPU: 58ms
   CGroup: /system.slice/apache2.service
           └─9964 /usr/sbin/apache2 -k start
             └─9967 /usr/sbin/apache2 -k start
               └─9968 /usr/sbin/apache2 -k start
                 └─9969 /usr/sbin/apache2 -k start
                   └─9970 /usr/sbin/apache2 -k start
                     └─9971 /usr/sbin/apache2 -k start

Oct 03 09:02:31 l systemd[1]: Starting apache2.service - The Apache HTTP Server...
Oct 03 09:02:31 l systemd[1]: Started apache2.service - The Apache HTTP Server.
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG#
```

Crearemos un archivo PHP con el comando **“echo """ | sudo tee /var/www/html/info.php “**. Y después verificaremos el archivo desde la terminal **“curl <http://localhost/info.php>”** y desde el navegador con **“<http://localhost/info.php>”**.

```
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# curl http://localhost/info.php
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml"><head>
<style type="text/css">
body {background-color: #fff; color: #222; font-family: sans-serif;}
pre {margin: 0; font-family: monospace;}
a:link {color: #009; text-decoration: none; background-color: #fff;}
a:hover {text-decoration: underline;}
table {border-collapse: collapse; border: 0; width: 934px; box-shadow: 1px 2px 3px rgba(0, 0, 0, 0.2);
}
.center {text-align: center;}
.center table {margin: 1em auto; text-align: left;}
.center th {text-align: center !important;}
td, th {border: 1px solid #666; font-size: 75%; vertical-align: baseline; padding: 4px 5px;}
th {position: sticky; top: 0; background: inherit;}
h1 {font-size: 150%;}
h2 {font-size: 125%;}
h2 a:link, h2 a:visited {color: inherit; background: inherit;}
.p {text-align: left;}
.e {background-color: #ccf; width: 300px; font-weight: bold;}
.h {background-color: #99c; font-weight: bold;}
.v {background-color: #ddd; max-width: 300px; overflow-x: auto; word-wrap: break-word;}
.v i {color: #999;}
img {float: right; border: 0;}
hr {width: 934px; background-color: #ccc; border: 0; height: 1px;}
:root {--php-dark-grey: #333; --php-dark-blue: #4F5B93; --php-medium-blue: #8892BF; --php-light-blue:
#E2E4EF; --php-accent-purple: #793862}@media (prefers-color-scheme: dark) {
body {background: var(--php-dark-grey); color: var(--php-light-blue)}
.h td, td.e, th {border-color: #606A90}
td {border-color: #505153}
.e {background-color: #404A77}
.h {background-color: var(--php-dark-blue)}
.v {background-color: var(--php-dark-grey)}
hr {background-color: #505153}
}
</style>
<title>PHP 8.3.6 - phpinfo()</title><meta name="ROBOTS" content="NOINDEX,NOFOLLOW,NOARCHIVE" /></head>
<body><div class="center">
```

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localhost/info.php

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PHP Version 8.3.6



System	Linux 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025 x86_64
Build Date	Jul 14 2025 18:30:55
Build System	Linux
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/8.3/apache2
Loaded Configuration File	/etc/php/8.3/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/8.3/apache2/conf.d
Additional .ini files parsed	/etc/php/8.3/apache2/conf.d/10-opcache.ini, /etc/php/8.3/apache2/conf.d/10-pdo.ini, /etc/php/8.3/apache2/conf.d/20-calendar.ini, /etc/php/8.3/apache2/conf.d/20-ctype.ini, /etc/php/8.3/apache2/conf.d/20-exif.ini, /etc/php/8.3/apache2/conf.d/20-ffi.ini, /etc/php/8.3/apache2/conf.d/20-fileinfo.ini, /etc/php/8.3/apache2/conf.d/20-ftp.ini, /etc/php/8.3/apache2/conf.d/20-gettext.ini, /etc/php/8.3/apache2/conf.d/20-iconv.ini, /etc/php/8.3/apache2/conf.d/20-ldap.ini, /etc/php/8.3/apache2/conf.d/20-odbc.ini, /etc/php/8.3/apache2/conf.d/20-oci8.ini, /etc/php/8.3/apache2/conf.d/20-pdo_sqlite.ini, /etc/php/8.3/apache2/conf.d/20-posix.ini, /etc/php/8.3/apache2/conf.d/20-readline.ini, /etc/php/8.3/apache2/conf.d/20-shmop.ini, /etc/php/8.3/apache2/conf.d/20-sockets.ini, /etc/php/8.3/apache2/conf.d/20-sysmsg.ini, /etc/php/8.3/apache2/conf.d/20-syssem.ini, /etc/php/8.3/apache2/conf.d/20-sysvshm.ini, /etc/php/8.3/apache2/conf.d/20-tokenizer.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	420230831
Zend Extension Build	API420230831.NTS
PHP Extension Build	API20230831.NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	disabled
Zend Max Execution Timers	disabled
IPv6 Support	enabled
DTrace Support	disabled
Registered PHP Streams	https, ftps, compress.zlib, php, file, glob, data, http, ftp, phar
Registered Stream Socket Transports	tcp, udp, unix, udg, ssl, tls, tlsv1.0, tlsv1.1, tlsv1.2, tlsv1.3
Registered Stream Filters	zlib*, string.rot13, string.toupper, string.tolower, convert*, consumed, dechunk, convert.iconv*

This program makes use of the Zend Scripting Language Engine:

Zend Engine v4.3.6, Copyright (c) Zend Technologies with Zend OPcache v8.3.6, Copyright (c), by Zend Technologies

zendengine

Configuration

Parte 2: Servidor Nginx con HTML

Primero pararemos el servicio de Apache con **“*sudo service apache2 stop*”**, luego instalaremos el servidor web Nginx con **“*sudo apt install nginx -y*”**, e iniciaremos el servicio con **“*sudo service nginx start*”**.

```
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# sudo service apache2 stop
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# |
```

```
</div></body></html>root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 564 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.5 [43
.4 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]
Fetched 564 kB in 0s (2090 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 41621 files and directories currently installed.)
Preparing to unpack ../nginx-common_1.24.0-2ubuntu7.5_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.5) ...
Selecting previously unselected package nginx.
Preparing to unpack ../nginx_1.24.0-2ubuntu7.5_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.5) ...
Setting up nginx-common (1.24.0-2ubuntu7.5) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/ng
inx.service.
Could not execute systemctl: at /usr/bin/deb-systemd-invoke line 148.
Setting up nginx (1.24.0-2ubuntu7.5) ...
Not attempting to start NGINX, port 80 is already in use.
Processing triggers for man-db (2.12.0-4build2) ...
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# |
```

```
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# sudo service nginx start
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# |
```

Para comprobar el funcionamiento de Nginx usaremos “***sudo systemctl status nginx***”.

```
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-10-03 09:13:34 CEST; 24s ago
     Docs: man:nginx(8)
   Process: 10375 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, >
   Process: 10377 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/>
 Main PID: 10378 (nginx)
    Tasks: 17 (limit: 9350)
   Memory: 12.7M (peak: 14.1M)
      CPU: 54ms
   CGroup: /system.slice/nginx.service
           └─10378 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─10379 "nginx: worker process"
               └─10380 "nginx: worker process"
                 └─10381 "nginx: worker process"
                   └─10382 "nginx: worker process"
                     └─10383 "nginx: worker process"
                       └─10384 "nginx: worker process"
                         └─10385 "nginx: worker process"
                           └─10386 "nginx: worker process"
                             └─10387 "nginx: worker process"
                               └─10388 "nginx: worker process"
                                 └─10390 "nginx: worker process"
                                   └─10391 "nginx: worker process"
                                     └─10392 "nginx: worker process"
                                       └─10393 "nginx: worker process"
                                         └─10394 "nginx: worker process"
                                           └─10395 "nginx: worker process"

Oct 03 09:13:34 l systemd[1]: Starting nginx.service - A high performance web server and a reverse pr>
Oct 03 09:13:34 l systemd[1]: Started nginx.service - A high performance web server and a reverse pro>
lines 1-31/31 (END)
```

Luego crearemos el archivo HTML con los comandos “***echo “<h1>Hola Mundo desde Nginx</h1><p>Servidor funcionando correctamente</p>”***”, y “***sudo tee /var/www/html/index.html***”.

```
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# echo "<h1>Hola Mundo desde Nginx</h1><p>Servidor funciona
do correctamente</p>"
<h1>Hola Mundo desde Nginx</h1><p>Servidor funcionando correctamente</p>
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG#
```

```
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# sudo tee /var/www/html/index.html
echo "<h1>Hola Mundo desde Nginx</h1><p>Servidor funcionando correctamente</p>"
echo "<h1>Hola Mundo desde Nginx</h1><p>Servidor funcionando correctamente</p>"
```

Por último, probaremos la página HTML en el navegador con “<http://localhost>”, y mostramos la IP para acceso desde Windows con “***ip addr show eth0 | grep inet***”.



```
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# ip addr show eth0 | grep inet
    inet 192.168.141.136/20 brd 192.168.143.255 scope global eth0
    inet6 fe80::215:5dff:fe84:f939/64 scope link
root@l:/mnt/c/Users/Alumno.DESKTOP-DI5KTUG# |
```